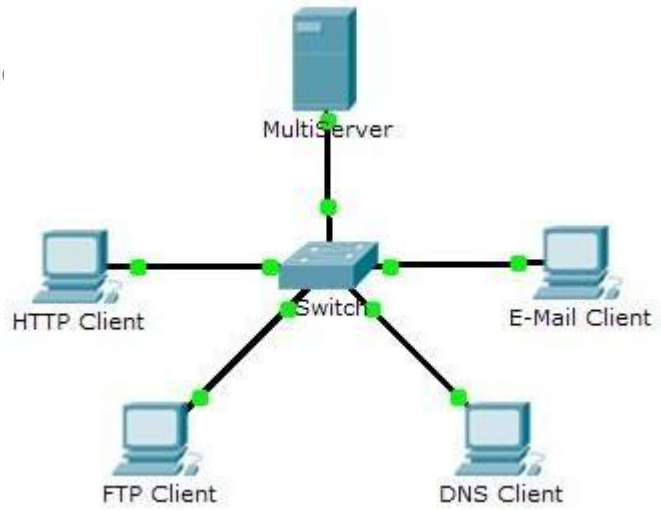
Packet Tracer Simulation - TCP and UDP Communications

Topology

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Objectives

Part 1: Generate Network Traffic in Simulation Mode

Part 2: Examine the Functionality of the TCP and UDP Protocols

Background

This simulation activity is intended to provide a foundation for understanding the TCP and UDP in detail. Simulation mode provides the ability to view the functionality of the different protocols.

As data moves through the network, it is broken down into smaller pieces and identified in some fashion so that the pieces can be put back together. Each of these pieces is assigned a specific name (protocol data unit [PDU]) and associated with a specific layer. Packet Tracer Simulation mode enables the user to view each of the protocols and the associated PDU. The steps outlined below lead the user through the process of requesting services using various applications available on a client PC.

This activity provides an opportunity to explore the functionality of the TCP and UDP protocols, multiplexing and the function of port numbers in determining which local application requested the data or is sending the data.

Part 1: Generate Network Traffic in Simulation Mode

Step 1: Generate traffic to populate Address Resolution Protocol (ARP) tables.

Perform the following tasks task to reduce the amount of network traffic viewed in the simulation. a.

Click **MultiServer** and click the **Desktop** tab > **Command Prompt**.

b. Enter the **ping 192.168.1.255** command. This will take a few seconds as every device on the network responds to **MultiServer**.

c. Close the **MultiServer** window.

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Step 2: Generate web (HTTP) traffic.

The HTTP client uses TCP protocol with destination port 80 to communicate with the web server.

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a. Switch to Simulation mode.

b. Click **HTTP Client** and click the **Desktop** tab > **Web Browser**.

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c. In the URL field, enter **192.168.1.254** and click **Go**. Envelopes (PDUs) will appear in the simulation window.

d. Minimize, but do not close, the **HTTP Client** configuration window.

Step 3: Generate FTP traffic.

a. Click **FTP Client** and click the **Desktop** tab > **Command Prompt**.

b. Enter the **ftp 192.168.1.254** command. PDUs will appear in the simulation window.

c. Minimize, but do not close, the **FTP Client** configuration window.

Step 4: Generate DNS traffic.

a. Click **DNS Client** and click the **Desktop** tab > **Command Prompt**.

b. Enter the **nslookup multiserver.pt.ptu** command. A PDU will appear in the simulation window.

c. Minimize, but do not close, the **DNS Client** configuration window.

Step 5: Generate Email traffic.

a. Click **E-Mail Client** and click the **Desktop** tab > **E Mail** tool.

b. Click **Compose** and enter the following information:

1) **To:** user@multiserver.pt.ptu
2) **Subject:** Personalize the subject line
3) **E-Mail Body:** Personalize the Email

c. Click **Send**.

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d. Minimize, but do not close, the **E-Mail Client** configuration window.

Step 6: Verify that the traffic is generated and ready for simulation.

Every client computer should have PDUs listed in the Simulation Panel.

Part 2: Examine Functionality of the TCP and UDP Protocols

Step 1: Examine multiplexing as all of the traffic crosses the network.

You will now use the **Capture/Forward button** and the **Back** button in the Simulation Panel. a.

Click **Capture/Forward** once. All of the PDUs are transferred to the switch.

b. Click **Capture/Forward** again. Some of the PDUs disappear. What do you think happened to them?

Some PDUs disappeared because they reached their destination successfully.

c. Click **Capture/Forward** six times. All clients should have received a reply. Note that only one PDU can cross a wire in each direction at any given time. What is this called?

This is called multiplexing.

d. A variety of PDUs appears in the event list in the upper right pane of the simulation window. Why are they so many different colors?

Different colors represent different protocols and different types of PDUs in the simulation.

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e. Click **Back** eight times. This should reset the simulation.

Note: Do not click **Reset Simulation** any time during this activity; if you do, you will need to repeat the steps in Part 1.

Step 2: Examine HTTP traffic as the clients communicate with the server.

a. Filter the traffic that is currently displayed to display only **HTTP** and **TCP** PDUs filter the traffic that is currently displayed:

1) Click **Edit Filters** and toggle the **Show All/None** check box.

2) Select **HTTP** and **TCP**. Click anywhere outside of the Edit Filters box to hide it. The Visible Events should now display only **HTTP** and **TCP** PDUs.

b. Click **Capture/Forward**. Hold your mouse above each PDU until you find one that originates from **HTTP Client**. Click the PDU envelope to open it.

c. Click the **Inbound PDU Details** tab and scroll down to the last section. What is the section labeled?

The section is labeled TCP.

Are these communications considered to be reliable?

Yes, TCP communications are considered reliable because they use acknowledgements and sequence numbers.

d. Record the **SRC PORT**, **DEST PORT**, **SEQUENCE NUM**, and **ACK NUM** values. What is written in the field to the left of the **WINDOW** field?

The field contains TCP flags such as SYN, ACK, or PSH.

e. Close the PDU and click **Capture/Forward** until a PDU returns to the **HTTP Client** with a checkmark.

f. Click the PDU envelope and select **Inbound PDU Details**. How are the port and sequence numbers different than before?

The source and destination ports are reversed, and the acknowledgement number increases.

g. There is a second **PDU** of a different color, which **HTTP Client** has prepared to send to **MultiServer**. This is the beginning of the HTTP communication. Click this second PDU envelope and select **Outbound PDU Details**.

h. What information is now listed in the TCP section? How are the port and sequence numbers different from the previous two PDUs?

The port numbers remain associated with the same applications, but the sequence and acknowledgement numbers continue increasing during communication.

i. Click **Back** until the simulation is reset.

Step 3: Examine FTP traffic as the clients communicate with the server.

a. In the Simulation Panel, change **Edit Filters** to display only **FTP** and **TCP**.

b. Click **Capture/Forward**. Hold your cursor above each PDU until you find one that originates from **FTP Client**. Click that PDU envelope to open it.

c. Click the **Inbound PDU Details** tab and scroll down to the last section. What is the section labeled?

The section is labeled TCP.

Are these communications considered to be reliable?

Yes, FTP uses TCP, which provides reliable communication.

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d. Record the **SRC PORT**, **DEST PORT**, **SEQUENCE NUM**, and **ACK NUM** values. What is written in the field to the left of the **WINDOW** field?

The field shows TCP flags such as SYN and ACK.

e. Close the PDU and click **Capture/Forward** until a PDU returns to the **FTP Client** with a checkmark.

f. Click the PDU envelope and select **Inbound PDU Details**. How are the port and sequence numbers different than before?

The source and destination ports are reversed and the acknowledgement number changes.

g. Click the **Outbound PDU Details** tab. How are the port and sequence numbers different from the previous two results?

The sequence and acknowledgement numbers continue to increase as the session progresses.

h. Close the PDU and click **Capture/Forward** until a second PDU returns to the **FTP Client**. The PDU is a different color.

i. Open the PDU and select **Inbound PDU Details**. Scroll down past the TCP section. What is the message from the server?

The server sends a welcome message for the FTP service.

j. Click **Back** until the simulation is reset.

Step 4: Examine DNS traffic as the clients communicate with the server.

a. In the Simulation Panel, change **Edit Filters** to display only **DNS** and **UDP**.

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b. Click the PDU envelope to open it.

c. Click the **Inbound PDU Details** tab and scroll down to the last section. What is the section labeled?

The section is labeled UDP.

Are these communications considered to be reliable? No, UDP communications are not considered reliable.

d. Record the **SRC PORT** and **DEST PORT** values. Why is there no sequence and acknowledgement number?

UDP does not establish connections and does not use sequence or acknowledgement numbers.

e. Close the **PDU** and click **Capture/Forward** until a PDU returns to the **DNS Client** with a checkmark.

f. Click the PDU envelope and select **Inbound PDU Details**. How are the port and sequence numbers different than before?

The source and destination ports are reversed. UDP does not use sequence numbers.

g. What is the last section of the **PDU** called?

The last section is called DNS.

h. Click **Back** until the simulation is reset.

Step 5: Examine email traffic as the clients communicate with the server.

a. In the Simulation Panel, change **Edit Filters** to display only **POP3, SMTP** and **TCP**.

b. Click **Capture/Forward**. Hold your cursor above each PDU until you find one that originates from **E-mail Client**. Click that PDU envelope to open it.

c. Click the **Inbound PDU Details** tab and scroll down to the last section. What transport layer protocol does email traffic use?

Email traffic uses TCP.

Are these communications considered to be reliable?

Yes, email communications are reliable because TCP guarantees delivery.

d. Record the **SRC PORT**, **DEST PORT**, **SEQUENCE NUM**, and **ACK NUM** values. What is written in the field to the left of the **WINDOW** field?

The field contains TCP flags such as SYN and ACK.

e. Close the **PDU** and click **Capture/Forward** until a PDU returns to the **E-Mail Client** with a checkmark.

f. Click the PDU envelope and select **Inbound PDU Details**. How are the port and sequence numbers different than before?

The source and destination ports are reversed and the acknowledgement numbers increase.

g. Click the **Outbound PDU Details** tab. How are the port and sequence numbers different from the previous two results?

The sequence and acknowledgement numbers continue increasing during communication.

h. There is a second **PDU** of a different color that **HTTP Client** has prepared to send to **MultiServer**. This is the beginning of the email communication. Click this second PDU envelope and select **Outbound PDU Details.**

i. How are the port and sequence numbers different from the previous two **PDU**s?

The port numbers identify different email services while the sequence numbers continue changing during the session.

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What email protocol is associated with TCP port 25? What protocol is associated with TCP port 110?

TCP port 25 is associated with SMTP.

TCP port 110 is associated with POP3.

j. Click **Back** until the simulation is reset.

Step 6: Examine the use of port numbers from the server.

The ping command was used to populate the ARP table, while the netstat command displayed active TCP and UDP connections on the server.

a. To see TCP active sessions, perform the following steps in quick succession:

1) Switch back to **Realtime** mode.

2) Click **MultiServer** and click the **Desktop** tab > **Command Prompt**.

b. Enter the **netstat** command. What protocols are listed in the left column?

What port numbers are being used by the server?

The protocols listed are TCP and UDP.

The server uses common port numbers such as 21 (FTP), 25 (SMTP), 53 (DNS), 80 (HTTP), and 110 (POP3).

c. What states are the sessions in?

The sessions are in states such as ESTABLISHED and TIME_WAIT.

d. Repeat the **netstat** command several times until you see only one session still ESTABLISHED. For which service is this connection still open?

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The FTP service connection remains open.

Why doesn't this session close like the other three? (Hint: Check the minimized clients)

The FTP session is open because the client is still connected and waiting for additional commands.

Suggested Scoring Rubric

Activity Section	Question Location	Possible Points	Earned Points
Part 2: Examine Functionality of the TCP and UDP Protocols	Step 1	15	
	Step 2	15	
	Step 3	15	
	Step 4	15	
	Step 5	15	
	Step 6	25	
	Total Score	**100**	

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